CIVIL ENGINEERING					
IV B.Tech – I Semester					
Course Code	**TRANSPORTATION ECONOMICS** **Professional Elective-V**	L	T	P	C
23A01703c		3	0	0	3

Course Objectives:

The objectives of this course are to make the student to:

1. Understand the fundamentals of transportation project development and decision-making.
2. Analyze transportation costs, including agency and user costs.
3. Evaluate vehicle operating costs and traffic congestion economics.
4. Apply economic evaluation methods for transportation projects.
5. Assess financing methods and risk analysisin transportation projects.

Course Outcomes (COs):

Upon successful completion of the course, students will be able to:

1. Describe the overall process of transportation project development and financial planning.
2. Analyze transportation cost structures, including demand and supply elasticity.
3. Evaluate vehicle operating costs and traffic congestion pricing strategies.
4. Apply economic analysis techniques to assess the feasibility of transportation projects.
5. Assess financial models, PPP strategies, and risk analysis for road projects.

CO – PO Articulation Matrix

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	3	2	2	1	2	-	-	-	-	2	-	2	2	2
CO -2	3	3	3	3	3	-	-	-	-	2	-	2	3	2
CO -3	3	3	3	3	2	-	-	-	-	2	-	2	3	3
CO -4	3	3	3	3	2	-	-	-	-	2	-	2	3	3
CO -5	3	3	3	3	3	3	3	3	-	2	-	2	3	3

UNIT – I

Introductory Conceptsin Transportation Decision Making: Overall Transportation Project Development, Budgeting, Financial Planning, The Process of Transportation Project Development, Models Associated With Transportation Impact Evaluation Professional Ethics.

UNIT – II

Transportation Costs - Classification of Transportation Costs, Transportation Agency Costs, Transportation User Costs, General Structure and Behavior of Cost Functions and Road Pricing. Estimating Transportation Demand and Supply - Supply Equilibration, Dynamics of Transportation Demand and Supply, Elasticity of Travel Demand and Supply, Classification of Elasticity

UNIT – III

Vehicle Operating Costs: Fuel Costs - Maintenance and Spares, Depreciation - Crew Costs -

Value of Travel Time Savings - Accident Costs. Economics of Traffic Congestion - Pricing Policy

UNIT – IV

Economic Analysis of Projects - Methods of Evaluation - Cost-Benefit Ratio, First Year Rate of Return, Net Present Value, and Internal-Rate of Return Methods; Indirect Costs and Benefits of Transport Projects

UNIT – V

Financing of Road Projects - Methods – Private Public Partnership (PPP) - toll Collection - Economic Viability of Design-Build-Operate-Transfer Schemes – Risk Analysis – Value for Money Analysis - Case Studies.

TEXT BOOKS:

1. Winfrey, Economic analysis for Highways, International Textbook Company, Pennsylvania, 1969.
2. Sarkar, P. K., andMaitri, V., Economicsin Highway and Transportation Planning, Standard Publisher, New Delhi, 2010.

REFRENCE BOOKS:

1. IRC, Manual on Economic Evaluation of Highway Projects in India, SP30, 2007
2. David, H., and Brewer, A., Transport: An Economics and Management Perspective. Oxford University Press, UK, 2000.
3. Quinet, E., andVickerman, R., Principles of Transport Economics, Edward Elgar Pub, 2005
4. Button, K. J., Transport Economics, Elgar, 2010

Online Learning Resources:

https://archive.nptel.ac.in/courses/105/104/105104098/
https://archive.nptel.ac.in/content/storage2/courses/105101087/01-Ltexhtml/p2/p.html